

<https://theconversation.com/can-big-data-really-predict-what-makes-a-song-popular-189052>

Music is part of our lives in different ways. We listen to it on our commutes and it resounds through shopping centres. Some of us seek live music at concerts, festivals and shows or rely on music to set the tone and mood of our days.

While we might understand the genres or songs we appreciate, it's not clear precisely why a certain song is more appealing or popular. Perhaps the lyrics speak to an experience? Perhaps the energy makes it appealing? These questions are important to answer for music industry professionals, and [analyzing data](#) is a key part of this.

At Carleton University, a group of data science researchers sought to answer the question: “What descriptive features of a song make it popular on music/online platforms?”

Revenue in the music industry

Revenue in the music industry [is derived from two sources that are affected by different factors: live music and recorded music](#). During the pandemic, although live music income dropped due to the cancellation of in-person performances, the [income from streaming](#) rose.

As digital platforms like Spotify and TikTok have grown, [the majority of music revenue has come to be contributed by digital media, mostly music streaming](#). How and whether this [revenue](#)

[reaches singers and songwriters at large](#) is another matter.

Popularity on digital platforms

The popularity of a song on digital platforms is considered a measure of the revenue the song may generate.

As such, producers seek to answer questions like “[How can we make the song more popular?](#)” and “[What are the characteristics of songs that make it the top charts?](#)”

With collaborators [Laura Colley](#), [Andrew Dybka](#), Adam Gauthier, Jacob Laboissonniere, Alexandre Mougeot and Nayeef Mowla, we produced a systematic study that collected data from YouTube, Twitter, TikTok, Spotify and Billboard ([Billboard Hot-100](#), sometimes also denoted by data researchers as “[Billboard hot top](#)” or in our work and others’ work, “Billboard Top-100”).

We linked the datasets from the different platforms with Spotify’s acoustic descriptive metric or “descriptive features” for songs. These features have been derived [from a dataset which yielded categories for measuring and analyzing qualities of songs](#). Spotify’s [metrics capture descriptive features such as](#) *acousticness*, *energy*, *danceability* and *instrumentalness* (the collection of instruments and voices in a given piece).

We sought to find trends and analyze the relationship between songs’ descriptive features and their popularity.

The rankings on the weekly [Billboard Hot-100](#) are based on sales, online streams and radio plays in the United States.

The analysis we performed by looking at Spotify and Billboard revealed insights that are useful for the music industry.